

Mohd Yousuf Parvez

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Bangalore, India

SUMMARY

Backend and AI Engineer with 3.5+ years of experience building Java microservices, agentic AI systems, and RAG pipelines for enterprise-scale applications. Experienced in LLM integration, prompt engineering, and AI output evaluation. Proven ability to diagnose distributed system bottlenecks using Dynatrace, design observable AI backends, and deliver production-ready services on AWS. Currently focused on AI Backend and MLOps engineering roles, with hands-on delivery across healthcare, logistics, and automotive domains.

TECHNICAL SKILLS

Languages: Java (Primary), Python, JavaScript, TypeScript, Bash / Shell Scripting

Backend & APIs: Spring Boot, Spring MVC, REST APIs, Microservices, Node.js, Express.js, Design Patterns

AI / LLM Engineering: LangChain, LangGraph, OpenAI API, Agentic AI Systems, RAG Pipelines, Prompt Engineering, LLM Evaluation, Hallucination Detection, Pinecone, FAISS, Weaviate, Embeddings

Observability & Performance: Dynatrace (APM, JVM profiling, distributed tracing), Prometheus, Grafana, Splunk, LoadRunner, JVM Heap & GC Analysis, AI Inference Monitoring

Databases: PostgreSQL, MySQL, Oracle, MongoDB, Couchbase, Elasticsearch

Cloud: AWS (EC2, S3, Lambda, CloudWatch), Microsoft Azure, GCP

DevOps: Docker, Kubernetes, Jenkins, Terraform, Git, CI/CD Pipelines, Linux Administration

Testing & QA: JUnit, Mockito, Cypress, Cucumber, AI Response Validation, Performance Regression Testing, TDD

EXPERIENCE

AutoZone / CHTS

Remote, TN

Performance Engineer & AI Backend Engineer

Jun 2025 – Present

- Designed and executed enterprise-scale load tests using LoadRunner and VGen simulating 10,000+ concurrent users across Java microservices and POS register fleets, ensuring SLA compliance under peak traffic.
- Built an AI anomaly detection agent using LangGraph and OpenAI API that autonomously ingests Dynatrace metric streams, classifies failure patterns, and generates root-cause summaries — reducing mean triage time by 40%.
- Developed a RAG-based internal knowledge assistant over 3,000+ pages of system runbooks and incident reports using LangChain, Pinecone, and OpenAI embeddings; integrated into Slack for on-call engineer self-service.
- Built an LLM evaluation harness to score AI-generated performance reports against ground-truth metrics, detecting hallucinations and output drift across model upgrades before reports reached stakeholders.
- Performed deep JVM heap analysis, GC profiling, and distributed trace inspection via Dynatrace to resolve memory leaks and thread contention; optimized Spring Boot API paths cutting P95 latency by 35%.
- Administered Linux-based environments for 100+ POS register nodes; authored Bash automation scripts for port lifecycle management, health checks, and multi-hour maintenance workflows across server fleets.
- Collaborated with DevOps to instrument AI and backend services with Dynatrace agents, deploy via Docker, and integrate performance regression gates into Jenkins CI/CD pipelines.
- Key Technologies:** Java, Spring Boot, Python, LangChain, LangGraph, OpenAI API, Pinecone, LoadRunner, Dynatrace, Docker, Jenkins, AWS, Linux, Bash

TheraPz (via VNC Digital Services)

Hyderabad, India

Software Engineer – Full Stack & DevOps

Dec 2024 – Apr 2025

- Built full-stack healthcare platform for autistic children's therapy management using React and Spring Boot microservices with PostgreSQL, supporting patient data, therapy tracking, and parent persona workflows.
- Designed and integrated an AI-powered clinical documentation assistant using LangChain and OpenAI API with RAG retrieval over vectorized patient history (FAISS), reducing therapist note time by 30%.
- Designed RESTful APIs for insurance integrations and billing workflows; implemented JWT-based authentication and RBAC across all service endpoints.
- Architected Jenkins and Docker CI/CD pipelines on AWS EC2 reducing release cycles from weekly to daily; set up Prometheus and Grafana dashboards monitoring service health against SLAs.

- **Key Technologies:** React, Spring Boot, PostgreSQL, LangChain, OpenAI API, FAISS, AWS EC2, Docker, Jenkins, Prometheus, Grafana

Cognizant Technology Solutions

Noida, India

Programmer Analyst – Backend & DevOps Engineer

Aug 2023 – Nov 2024

- Developed and owned backend microservices for enterprise clients (Dell EMC UK, Maersk Air Cargo) using Java Spring Boot, MongoDB, and PostgreSQL, handling millions of daily transactions at sub-100ms P99 latency.
- Automated CI/CD pipelines for 10+ microservices using Jenkins, Maven, and Ansible; enforced code quality gates via SonarQube maintaining 95%+ coverage across all owned services.
- Containerized services with Docker and orchestrated zero-downtime deployments using Kubernetes on AWS and OpenShift; managed canary and rolling release strategies for production services.
- Implemented Splunk and Prometheus monitoring with custom alerting; provided rotational on-call support resolving production incidents within SLA windows.
- **Key Technologies:** Java, Spring Boot, MongoDB, PostgreSQL, Jenkins, Docker, Kubernetes, OpenShift, AWS, Splunk, Ansible, SonarQube

Cognizant Technology Solutions

Noida, India

Programmer Analyst – Full Stack Engineer

Aug 2022 – Jul 2023

- Built full-stack enterprise applications using Spring Boot, Angular, and MySQL; designed robust backend REST APIs and integrated Angular frontends across multiple business domains.
- Administered Linux application server environments — managed service lifecycles, log analysis, cron scheduling, and wrote Bash scripts for deployment and operational automation.
- Established Git and Jenkins-based CI/CD workflows with Maven; achieved 95%+ unit test coverage using JUnit and TDD methodology across all owned services.
- Provisioned and managed AWS EC2 instances; containerized applications using Docker for consistent cross-environment deployments.
- **Key Technologies:** Java, Spring Boot, Angular, MySQL, Linux, Bash, Jenkins, Git, Maven, AWS EC2, Docker, JUnit

PROJECTS

Autozone Autonomous Incident Response Agent

LangGraph, LangChain, OpenAI API, Pinecone, FastAPI, PostgreSQL, Docker

- Built a multi-step agentic AI system using LangGraph that autonomously monitors system health metrics, classifies anomalies, queries a RAG knowledge base of past incident runbooks, and drafts resolution steps.
- Implemented a Pinecone-backed retrieval layer over 5,000+ historical incident reports and runbook documents; agent achieves 87% accuracy matching new incidents to known resolution patterns.
- Exposed agent via FastAPI with async streaming (SSE), integrated into Slack via webhook; reduced on-call engineer first-response time from 18 minutes to under 4 minutes in internal testing.

AZ Close LLM Evaluation & QA Framework | *Open Source*

Python, OpenAI API, Playwright, FastAPI, PostgreSQL, React

- Built a full-stack evaluation framework for LLM-powered applications covering prompt regression testing, hallucination detection, schema validation, and response consistency scoring across model versions.
- Implemented headless test runners using Playwright to validate AI outputs against expected business rules on non-deterministic API responses; supports parallel execution across 50+ test cases.
- Built a React dashboard tracking eval pass rates, latency trends, and quality drift per model version — enabling safe prompt iteration and zero-surprise model upgrades in production.

AZ Stores RAG-Powered Enterprise Knowledge Assistant

LangChain, OpenAI API, Weaviate, Spring Boot, React

- Designed and deployed a RAG pipeline ingesting internal documentation (PDFs, Confluence, Notion exports) into a Weaviate vector store with OpenAI embeddings, enabling natural language Q&A over company knowledge.
- Built a Spring Boot API layer with semantic search, query rewriting, and citation tracking; React frontend with conversation history and source attribution per answer.
- Achieved sub-400ms end-to-end retrieval on a 80,000-chunk corpus; implemented query caching and re-ranking to reduce OpenAI token usage by 35%.

Caterpillar South India – Heavy Equipment Telemetry System

React, Spring Boot, MySQL, Terraform, Jenkins, AWS

- Built a real-time telemetry platform ingesting sensor data from 10,000+ industrial machines daily with operational analytics dashboards and automated alerting pipelines.
- Provisioned AWS infrastructure using Terraform; achieved 99.99% uptime with monitoring and disaster recovery protocols for mission-critical logistics operations.

Maersk Air Cargo – Global Logistics and Tracking Platform

Angular, Spring Boot, PostgreSQL, Docker, Jenkins, AWS

- Developed cargo tracking, route optimization, and real-time status update APIs for Maersk Air Cargo UK, serving global air freight operations across 40+ countries.
- Containerized and deployed via Docker and Jenkins CI/CD on AWS; integrated real-time event streaming for shipment status propagation to downstream systems.

EDUCATION

Shri Ramswaroop Memorial University (SRMU)

Lucknow, India

B.Tech – Computer Science, Big Data Specialization

Jun 2022

- Final year project *Crypto Pal* ranked in top 3 projects of the graduating cohort.

CERTIFICATIONS

Cloud: AWS Certified Cloud Practitioner | Azure Fundamentals (AZ-900) | Azure Data Fundamentals (DP-900) | OCI Foundations Associate | OCI AI Foundations Associate | OCI Data Foundations Associate

Development & AI: Core Java 8 (Intermediate) | Generative AI Fundamentals | AWS CodeWhisperer | Agile Software Development